

## SEMESTER 2 EXAMINATIONS 2020/2021

|                       |                                                                                                                                                         |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MODULE:</b>        | CA208 - Logic                                                                                                                                           |
| <b>PROGRAMME(S):</b>  | CASE<br>ECSAO                                                                                                                                           |
|                       | BSc in Computer Applications (Sft.Eng.)<br>Study Abroad (Engineering & Computing).                                                                      |
| <b>YEAR OF STUDY:</b> | 2,O                                                                                                                                                     |
| <b>EXAMINERS:</b>     | Dr. David Sinclair      (Internal)      (Ext:5510)                                                                                                      |
| <b>TIME ALLOWED:</b>  | 2 hours                                                                                                                                                 |
| <b>INSTRUCTIONS:</b>  | Answer <u>ALL</u> 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks. |

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**PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO**

The use of programmable or text storing calculators is expressly forbidden.  
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

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*There are no additional requirements for this paper.*

## Section A

### QUESTION 1

**[Total marks: 30]**

Consider the following problem.

- Either taxes are increased or if expenditures rise then the debt ceiling is raised.
- If taxes are increased, then the cost of collecting taxes rises.
- If a rise in expenditures implies that the government borrows more money, then if the debt ceiling is raised, then interest rates increase.
- If taxes are not increased and the cost of collecting taxes does not increase then if the debt ceiling is raised, then the government borrows more money.
- The cost of collecting taxes does not increase.
- Either interest rates do not increase or the government does not borrow more money.

1(a)

[10 Marks]

Write the problem above as a set of propositional formulae.

1(b)

[10 Marks]

Convert the propositional formulae from part (a) into Conjunctive Normal Form.

1(c)

[10 Marks]

Using *Resolution*, prove that either the debt ceiling isn't raised or expenditures don't rise.

**[End Question 1]**

### QUESTION 2

**[Total marks: 10]**

[10 Marks]

Without using truth tables, show that

$$(p \rightarrow q) \wedge \neg q \equiv \neg(p \vee q)$$

Clearly indicate which equivalence is used at each step.

**[End Question 2]**

**QUESTION 3****[Total marks: 10]**

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$\vdash ((p \wedge q) \rightarrow r) \rightarrow ((p \rightarrow r) \vee (q \rightarrow r))$$

Clearly indicate which rule was applied at each step.

**[End Question 3]****QUESTION 4****[Total marks: 10]**

[10 Marks]

Using the Gentzen System  $\mathcal{G}$  prove

$$\vdash (A \rightarrow B) \rightarrow ((\neg A \rightarrow B) \rightarrow B)$$

Clearly indicate which rule was applied at each step.

**[End Question 4]****QUESTION 5****[Total marks: 10]**

[10 Marks]

Using *semantic tableaux* prove the following first-order statement is valid.

$$\vdash (\forall x.p(x) \vee \forall x.q(x)) \rightarrow \forall x.(p(x) \vee q(x))$$

Clearly indicate which rule was applied at each step.

**[End Question 5]****QUESTION 6****[Total marks: 10]**

[10 Marks]

Using Gentzen system  $\mathcal{G}$ , prove the following predicate formula.

$$\forall x.(p(x) \rightarrow q(x)) \rightarrow (\forall x.p(x) \rightarrow \forall x.q(x))$$

Clearly indicate which rule was applied at each step.

**[End Question 6]**

## Section B

### QUESTION 7

**[Total marks: 10]**

[10 Marks]

Let a set be represented by a list of numbers. Write the following Prolog relations:

- `union(A,B,C)` is true if the list `C` contains only even numbers in either lists `A` or `B`.
- `intersection(A,B,C)` is true if the list `C` contains only even numbers in both lists `A` and `B`.

**[End Question 7]**

### QUESTION 8

**[Total marks: 10]**

[10 Marks]

Let a bus timetable be represented by a list of routes where each route is represented by `Start/End/DepartureTime/ArrivalTime`. Both `DepartureTime` and `ArrivalTime` are represented by a relation `time(Hour,Minutes)`. Write a Prolog relation `journey(Start, End)` that is true if there is a route in the timetable that can get a passenger from `Start` to `End` directly or any number of intermediate stops where there is at least 10 minutes between the arrival at an intermediate point before the departure of the next leg of the journey. Assume the clock is a 24 hour clock and all departure and arrival times occur within the same day.

**[End Question 8]**

### QUESTION 9

**[Total marks: 10]**

[10 Marks]

Describe *cuts* in Prolog. Why are *cuts* used? Illustrate your answer with examples.

**[End Question 9]**

## APPENDICES

### Semantic Tableaux

| $\alpha$                    | $\alpha_1$            | $\alpha_2$            | $\beta$                         | $\beta_1$                   | $\beta_2$                   |
|-----------------------------|-----------------------|-----------------------|---------------------------------|-----------------------------|-----------------------------|
| $\neg\neg A_1$              | $A_1$                 |                       | $\neg(B_1 \wedge B_2)$          | $\neg B_1$                  | $\neg B_2$                  |
| $A_1 \wedge A_2$            | $A_1$                 | $A_2$                 | $B_1 \vee B_2$                  | $B_1$                       | $B_2$                       |
| $\neg(A_1 \vee A_2)$        | $\neg A_1$            | $\neg A_2$            | $B_1 \rightarrow B_2$           | $\neg B_1$                  | $B_2$                       |
| $\neg(A_1 \rightarrow A_2)$ | $A_1$                 | $\neg A_2$            | $B_1 \uparrow B_2$              | $\neg B_1$                  | $\neg B_2$                  |
| $\neg(A_1 \uparrow A_2)$    | $A_1$                 | $A_2$                 | $\neg(B_1 \downarrow B_2)$      | $B_1$                       | $B_2$                       |
| $A_1 \downarrow A_2$        | $\neg A_1$            | $\neg A_2$            | $\neg(B_1 \leftrightarrow B_2)$ | $\neg(B_1 \rightarrow B_2)$ | $\neg(B_2 \rightarrow B_1)$ |
| $A_1 \leftrightarrow A_2$   | $A_1 \rightarrow A_2$ | $A_2 \rightarrow A_1$ | $B_1 \oplus B_2$                | $\neg(B_1 \rightarrow B_2)$ | $\neg(B_2 \rightarrow B_1)$ |
| $\neg(A_1 \oplus A_2)$      | $A_1 \rightarrow A_2$ | $A_2 \rightarrow A_1$ |                                 |                             |                             |

  

| $\gamma$             | $\gamma(a)$ | $\delta$             | $\delta(a)$ |
|----------------------|-------------|----------------------|-------------|
| $\forall x.A(x)$     | $A(a)$      | $\exists x.A(x)$     | $A(a)$      |
| $\neg\exists x.A(x)$ | $\neg A(a)$ | $\neg\forall x.A(x)$ | $\neg A(a)$ |

### Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$  is a *sequent* and means that if  $A$  is true, then  $B$  can be inferred.

| $\alpha$                        | $\alpha_1$                  | $\alpha_2$                  | $\beta$                     | $\beta_1$             | $\beta_2$             |
|---------------------------------|-----------------------------|-----------------------------|-----------------------------|-----------------------|-----------------------|
| $\neg\neg A$                    | $A$                         |                             | $B_1 \wedge B_2$            | $B_1$                 | $B_2$                 |
| $\neg(A_1 \wedge A_2)$          | $\neg A_1$                  | $\neg A_2$                  | $\neg(B_1 \vee B_2)$        | $\neg B_1$            | $\neg B_2$            |
| $A_1 \vee A_2$                  | $A_1$                       | $A_2$                       | $\neg(B_1 \rightarrow B_2)$ | $B_1$                 | $\neg B_2$            |
| $A_1 \rightarrow A_2$           | $\neg A_1$                  | $A_2$                       | $\neg(B_1 \uparrow B_2)$    | $B_1$                 | $B_2$                 |
| $A_1 \uparrow A_2$              | $\neg A_1$                  | $\neg A_2$                  | $B_1 \downarrow B_2$        | $\neg B_1$            | $\neg B_2$            |
| $\neg(A_1 \downarrow A_2)$      | $A_1$                       | $A_2$                       | $B_1 \leftrightarrow B_2$   | $B_1 \rightarrow B_2$ | $B_2 \rightarrow B_1$ |
| $\neg(A_1 \leftrightarrow A_2)$ | $\neg(A_1 \rightarrow A_2)$ | $\neg(A_2 \rightarrow A_1)$ | $\neg(B_1 \oplus B_2)$      | $B_1 \rightarrow B_2$ | $B_2 \rightarrow B_1$ |
| $A_1 \oplus A_2$                | $\neg(A_1 \rightarrow A_2)$ | $\neg(A_2 \rightarrow A_1)$ |                             |                       |                       |

  

| $\gamma$             | $\gamma(a)$ | $\delta$             | $\delta(a)$ |
|----------------------|-------------|----------------------|-------------|
| $\exists x.A(x)$     | $A(a)$      | $\forall x.A(x)$     | $A(a)$      |
| $\neg\forall x.A(x)$ | $\neg A(a)$ | $\neg\exists x.A(x)$ | $\neg A(a)$ |

**[END OF APPENDICES]**

**[END OF EXAM]**